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Optimal Lesion Size Index to Prevent Conduction Gap during Pulmonary Vein Isolation

Short title: Optimal Lesion Size Index in Pulmonary Vein Isolation

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This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1111/jce.13727.

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S. Sakagami reports travel support and compensation for speaking on this topic, but did not specify the sponsor. Other authors: No disclosures.

Abstract

Introduction:

A novel real-time lesion size index (LSI) that incorporates contact force (CF), time, and power has been developed for safe and effective catheter ablation. We evaluated the optimal LSI to eliminate gap formation during pulmonary vein isolation (PVI).

Methods and Results:

We enrolled consecutive patients who underwent their first PVI using fiber-optic CF-sensing catheter for atrial fibrillation between December 2016 and October 2017. CF parameters, force–time integral (FTI), and LSI for 3,095 ablation points in 34 patients were evaluated. FTI and LSI in the lesions with gaps or dormant conduction (gaps/DC) were significantly lower than those in the lesion without gaps/DC [FTI: 140.5 ± 54.5 and 232.4 ± 121.4 g･s ($p < 0.0001$); LSI: 4.0 ± 0.6 and 4.7 ± 0.9 ($p < 0.0001$), respectively]. On receiver operating characteristic curve analysis, the optimal LSI threshold was 4.05 (sensitivity 63.4%; specificity 76.3%). LSI of <5.25 predicted gap or DC with a high sensitivity (sensitivity 97.6%; specificity 25.7%). In the posterior wall, which was 37% thinner than the non-posterior wall, a lower LSI of <3.95 showed relatively high sensitivity (92.3%) and specificity (65.6%).

Conclusions:

LSI can be used to predict gaps/DC during PVI procedure. LSI ≈ 5.2 may be a suitable target for effective lesion formation. LSI ≈ 4.0 may be acceptable in the posterior wall, especially in areas adjacent to the esophagus.

Keywords: catheter ablation; atrial fibrillation; contact force; lesion size index; pulmonary vein isolation

Introduction

The efficacy and safety of radiofrequency (RF) catheter ablation for atrial fibrillation (AF) have been established with procedural and technical advances.[1] Lesion formation is traditionally monitored as electrogram diminution, impedance drop, and changes in tip electrode temperature during RF application.[2,3] However, the accuracy of those surrogate measures has not been extensively validated, and these measures do not provide information of electrode-to-tissue interaction. Several *in vivo* and *in vitro* studies have emphasized the relationship between electrode–tissue contact and lesion formation.[2,4] Clinical studies that used electrode-to-tissue contact force (CF)-sensing catheters revealed CF as a major determinant of the generated lesion size.[2,3,5-7] Recently, a novel ablation catheter (TactiCath Quartz; St. Jude Medical, USA) has been developed to measure real-time catheter–tissue CF using the optical fibers during catheter mapping and RF ablation. This CF-sensing catheter offers the opportunity to obtain ablation indices, such as force–time integral (FTI). Several studies have demonstrated a correlation between FTI and lesion size during RF application;[8] further, FTI was shown to predict lesion size and steam pop occurrence in beating canine heart.[9] The lesion size index (LSI) is another novel multiparametric index that incorporates time, power, CF, and impedance data recorded during RF ablation in a weighted formula that describes ablation biophysics.[10,11] However, optimal LSI for successful Pulmonary vein isolation (PVI) procedure has not been verified. The aim of this study was to evaluate the optimal LSI for anatomical PVI procedure using TactiCath Quartz.

Methods

Study population

Thirty-four consecutive patients with drug-refractory AF who underwent their first PVI using fiber-optic CF sensing catheter (TactiCath Quartz) between December 2016 and October 2017 were enrolled in this study. All patients were administered an anticoagulant for at least >3 weeks immediately prior to the procedure. The absence of thrombus in the left atrium (LA) was confirmed by pre-procedural laboratory investigations and cardiac computed tomography (CT) angiogram or transesophageal echocardiogram.

This study complied with the Declaration of Helsinki, and was approved by the ethics committee of the National Hospital Organization, Kanazawa Medical Center. Written informed consent was obtained from all patients prior to the procedure.

Pulmonary vein isolation procedure

PVI was achieved with a wide antrum ablation line around the pulmonary veins (PVs) guided by a three-dimensional mapping system (EnSite Precision; St. Jude Medical). The procedure was performed under general anesthesia with propofol. A multielectrode catheter (BeeAT; Japan Lifeline, Japan) was transvenously inserted and positioned in the coronary sinus. An 8F intracardiac echocardiography catheter (64-element, 4.0–10.0 MHz, AcuNav; Biosense Webster, USA) was advanced into the right atrium. After a single transseptal puncture was performed under intracardiac ultrasound guidance, unfractionated heparin was intravenously administered to maintain an activated clotting time of >300 s. Through transseptal access, nonsteerable sheath (Swartz Braided Transseptal Guiding Introducer SL0; St. Jude Medical) and steerable sheath (Agilis; St. Jude Medical) were placed into the LA. Before ablation, three-dimensional anatomical model creation and electrical mapping of the LA and PVs were performed using the circular catheter (Advisor FL Circular Mapping Catheter, Sensor Enabled; St. Jude Medical), which was advanced into the LA via a steerable sheath, with reference to the reconstructed three-dimensional CT data. The irrigated CF-sensing ablation catheter (TactiCath Quartz) was advanced into the LA through a steerable sheath, and extensive encircling PVI was performed, guided by the three-dimensional anatomical model (Figure 1A). Ablation was performed in a temperature-controlled mode with the temperature limited to 43°C and the power limited to 25 W in the posterior segments, 35 W in left lateral ridge segments, and 30 W in the other segments. The irrigation flow rate was preset at 2 mL/min during mapping and at 17–30 mL/min during ablation. The target CF was between 10 and 30 g during PVI. CF and FTI were not blinded to the operator, whereas LSI was blinded. Serial ablation was performed with continuous dragging or point by point except for the regions of the posterior wall in contact with the esophagus. The timing of proceeding from one ablation site to the next was left to the each operator's judgement based on electrogram abatement, FTI (≥ 200 g・s was generally targeted), and increase in luminal esophageal temperature above 39°C; which was monitored using a luminal esophageal temperature probe (SensiTherm; St. Jude Medical). We set the ablation tag diameter at 4 mm and intended to ablate so that a new tag overlapped with the former one, and consequently the targeted inter-lesion distance was 3 to 4 mm (Figure 1B). The circumferential isolation was first performed anatomically without the circular catheter, using an exclusively anatomical approach. PV antrum isolation (defined as absence of any PV potential or LA potential in the antral ablation area) was verified using the circular catheter and/or the ablation catheter electrogram. The patients who were in AF rhythm at the beginning of the procedure underwent PV antrum ablation during AF rhythm, and then were cardioverted to verify the conduction block. The decision to perform additional linear ablation or ablation of complex fragmented electrograms depended on the operator; however, the data pertaining to those additional ablation points were excluded from the analysis.

AutoMark settings

In this study, PVI was performed with the AutoMark system (St. Jude Medical). AutoMark system is an automated system that records the ablation points on 3D mapping. To characterize the effect of CF applied over time, the system automatically detects the ablation time and calculates FTI and LSI only when the ablation catheter stays within the confined area. FTI was defined as the total CF integrated over the time of RF delivery, and the formula of LSI was derived from a mathematical expression that incorporates CF, impedance, power, and duration of RF application.[11] AutoMark settings for filter thresholds in this study were as follows. For catheter position stability, the minimum marker time was 3 s, marker spacing was 3 mm, and the away time was 8 s.

Contact force analysis

Ablation points were assigned to the respective segments of the PV antrum. FTI, LSI, maximum CF, average CF, RF duration, and RF power were obtained for each ablation point. The site of RF application was recorded relative to eight predefined circumferential regions around each ipsilateral pair of PVs. The posterior wall of the LA was defined as the regions including posterior superior, posterior middle, or posterior inferior wall, whereas non-posterior consisted of five other regions.

Evaluation of residual conduction gaps and dormant conductions

After anatomical PV antrum isolation, all PVs were mapped to detect PV-to-LA conduction gaps. In the absence of gaps, the presence of dormant conduction (DC) was assessed in each PV after intravenous administration of 20 mg adenosine. The positions where additional RF application resulted in a change of sequence or abolition of the PV potential were defined as gap sites. DC was defined as reappearance of PV conduction (PV spikes of more than one beat) as recorded by the circular catheter. The positions where RF application resulted in a change of sequence or abolition of DC were defined as DC sites. When one gap or DC site included multiple ablation points with the smallest tag size, all of these points were counted as gaps or DCs.

CT data analysis

Contrast-enhanced CT scanning was performed using a 64-detector-row CT [multidetector CT (MDCT)] scanner (Brilliance CT; Philips Electronics, The Netherlands). The data acquisition window was set at 40%–75% of the R-R interval. Volume data were reconstructed into axial images (slice thickness of 0.9 mm) and transferred to a workstation for post-processing (Ziostation2; Ziosoft, Japan). For each PV, we divided the PV antrum into eight segments under the ablation line, and quantitative measurements for each region were performed, as reported previously.[12]

Statistical analysis

All analyses were performed using SPSS for Windows version 25.0 software (IBM, USA). The level of statistical significance was set at a two-sided p-value < 0.05. Data pertaining to patient characteristics are presented as mean ± standard deviation or percentage. The predictive value of different threshold levels of FTI and LSI for detection of gaps or DC was assessed using receiver operating characteristic (ROC) curve analysis.

Results

Characteristics of the study population and results of ablation

The characteristics of the study population are summarized in Table 1. The study population [n = 34; mean age: 68.4 ± 6.8 years; males 26 (76.5%)] included 18 patients with paroxysmal AF and 16 patients with persistent AF. The average left ventricular ejection fraction and left atrial diameter were 62.4 ± 12.4% and 40.3 ± 5.7 mm, respectively. In all 34 patients, PVs were successfully isolated by anatomical PV antrum ablation. The total ablation points with valid LSI in the entire cohort were 3,095 (average: 91.0 ± 14.6 ablation points per patient). The number of ablation points in the posterior wall and non-posterior wall were 979 and 2,116, respectively. Additional linear ablation or ablation of complex fragmented electrograms was performed for 16 (47.1%) patients. Mean operation time (defined as the time from puncture to removal of the sheath) was 112.8 ± 33.0 min. No major complications were encountered during and after the PVI procedure.

Myocardial wall thickness of the LA under the RF ablation line

Myocardial wall thickness of the LA under the ablation line was measured using MDCT (Figure 2). The mean wall thickness was 3.2 ± 1.2 mm. The LA posterior wall was 37% thinner than that of the non-posterior wall (2.3 ± 0.4 and 3.6 ± 1.3 mm, respectively, $p < 0.0001$).

CF parameters for each segment under the ablation line

Among the 3,095 ablation pointes in this study, 41 gaps were identified, whereas no DCs were observed. The CF parameters with and without gaps or DCs are shown in Table 2. Although the average power was not significantly different between the two groups, the max CF and average CF were lower, and the ablation time for each point was shorter in the lesion with gaps or DCs as compared with that in lesions without gaps or DCs. In addition, FTI and LSI in lesions with gaps or DCs were significantly lower than those in lesions without gaps or DCs. Figure 3 shows the distribution of FTI and LSI in each segment of the PVs. The FTI showed large variation in each PVs segment; the highest FTI was in the right anterior carina (289.5 ± 149.3 g･s), and the lowest FTI was in the left posterior middle segment (192.7 ± 82.9 g･s) where the esophagus is generally in close proximity. In contrast, the LSI showed relatively uniform value across each PV segment. A significant correlation between FTI and LSI was observed under different power settings as shown in Figure 4 [power setting of 25 W ($R^2 = 0.453$, $p < 0.0001$); 30 W ($R^2 = 0.548$, $p < 0.0001$); and 35 W ($R^2 = 0.413$, $p < 0.0001$)].

Prediction of gaps and DCs after anatomical PV antrum ablation

The distribution of FTI and LSI with and without gaps or DCs is shown in Figure 5. Of note, the distribution of gaps and DCs in the posterior wall was concentrated in low LSI of <4.0 in contrast to that in the non-posterior wall. ROC curves were made for FTI, LSI, average CF, max CF, and duration of RF application to determine the thresholds that predict the presence of gaps and DCs (Figure 6). The areas under the curve for FTI and LSI for all points were not significantly different; 0.774 (95% CI 0.730–0.819) and 0.724 (95% CI 0.659–0.790), respectively ($p = 0.21$). The optimal LSI threshold for predicting gaps or DCs was 4.05 (sensitivity 63.4%; specificity 76.3%). LSI of <5.25 highly predicted gap or DC formation (sensitivity 97.6%;

specificity 25.7%). In the posterior wall, the lower LSI of <3.95 showed relatively high sensitivity of 92.3% and specificity of 65.6%.

Influences of atrial rhythm on CF parameters

Nineteen patients (55.9%) underwent PV antrum ablation during AF rhythm with their ablation points of 1,806. Although average CF, max CF, and FTI were not significantly different between the patients in AF rhythm and those in sinus rhythm (average CF: 16.0 ± 7.0 vs 16.0 ± 7.3 g, $p = 0.661$; max CF: 35.2 ± 16.6 vs 36.4 ± 16.8 g, $p = 0.052$; FTI: 229.8 ± 124.0 and 233.0 ± 117.3 g･s, $p = 0.472$, respectively), LSI was lower in patients with AF rhythm than in those with sinus rhythm (4.6 ± 0.9 and 4.8 ± 0.9, respectively, $p < 0.0001$). The optimal LSI threshold for predicting gaps or DCs was 4.05 (sensitivity 69.6%; specificity 73.0%) in AF and 4.15 (sensitivity 61.1%; specificity 77.6%) in sinus rhythm.

Complications

In this study, no complications related to the ablation procedure such as audible pop, perforation of atrial wall, cardiac tamponade, atrio-esophageal fistula, phrenic nerve injury, or stroke were observed.

Discussion

We evaluated the utility of an optimal value for LSI to predict residual conduction gaps or DCs after anatomical PV antrum ablation. Our major findings were as follows: (1) low LSI was significantly associated with the formation of conduction gaps or DCs; (2) use of LSI threshold of <5.25 to predict gap or DC formation was associated with a high sensitivity (97.6%); (3) for the posterior wall, which is 37% thinner than the non-posterior wall, a lower LSI threshold of <3.95 showed an acceptable sensitivity (92.3%).

Low average CF has been reported as a strong predictor of residual gap in PVI procedure. In the TactiCath Contact Force Ablation Catheter Study for Atrial Fibrillation study, CF higher than 10 g was associated with increased treatment success.[13] FTI is defined as the total area under the CF curve applied during RF ablation; it has been used to predict lesion size and durability using CF.[8] FTI is derived from a simple multiplication of CF with time and does not take into account the delivered power that may be a strong predictor of lesion size. LSI is a novel index that can help predict the size of the lesion created by RF application. LSI is derived from a mathematical expression that incorporates CF, impedance, power, and duration of RF application.[11] LSI represents an improvement over FTI in several respects.[10] Firstly, LSI incorporates the energization parameter directly. Secondly, LSI reflects the shifting pattern of heating that is the shift from resistive (joule) heating during the early phase of lesion formation to diffusive (conductive) heating during the later phase. Thirdly, the delay between the variation of force and/or current and the change in lesion growth rate due to thermal latency and the fact that lesions rapidly grow to a certain depth (typically about 3 mm), beyond which the depth parameter continues to grow at a slower rate, are well-recognized phenomenon during the development process of LSI. There are some previous reports for the optimal FTI in PVI. To create transmural lesions or to avoid reconnection, FTI of 400 g・s is necessary in PVI procedure.[14,15] However, to the best of our knowledge, the role of LSI in AF ablation is not well characterized, and the optimal LSI in PVI procedure remains an open question.

In this study, we explored the effect of LSI in PVI procedure and found that low LSI was significantly associated with the formation of conduction gaps or DCs. Also, we evaluated the optimal threshold of LSI for predicting gaps or DCs after PVI. On ROC curve analysis, LSI threshold level of 4.05 to predict gaps or DCs was associated with a modest sensitivity (63.4%). However, considering that even a few remaining gaps or DCs from the ablation points could lead to AF recurrence, LSI of 5.25 with higher sensitivity of 97.6% could be a more suitable target to avoid residual gaps or DCs.

Esophageal injury and atrio-esophageal fistula are serious complications of PVI procedure that are associated with high mortality rates. Although the exact mechanisms of development of esophageal complications are not clear, the formation of atrio-esophageal fistula is related to time-dependent heating of the esophagus.[16] Therefore, minimal RF application to the LA posterior wall adjacent to the esophagus is recommended. In the previous study, we have demonstrated that atrial wall thickness is not uniform under the ablation line of PVI and that optimal FTI varies at each point.[12] In this study, the thickness of the posterior wall was actually 37% thinner compared with that of the non-posterior wall. Therefore, we sought to perform subanalysis of optimal LSI for posterior wall to avoid esophageal injury and found that LSI of <3.95 showed relatively high sensitivity for detecting residual gaps or DCs. On the basis of these results, we propose LSI of 4.0 as an acceptable target to avoid the formation of residual gaps or DCs in the posterior wall where RF ablation potentially causes esophageal injury.

Study limitations

In an *in vitro* model, the predictive value of LSI for lesion size was reported to be higher than that of FTI[11]; however, the superiority of LSI for predicting residual gap or DC was not validated in our human study. This discrepancy may possibly be explained by several factors. Firstly, in this study, the operators performed RF application with real-time monitoring of FTI, while they were blinded to LSI, which is a clear and strong bias. Secondly, the stability of the catheter is liable to fluctuate because of changes in respiratory pattern,[17] atrial rhythm,[18] or anatomical situation[15,19,20] in humans. Thirdly, we performed RF application with preset power setting of 20–35 W according to the regions. For these reasons, this study is not suitable for comparing the usefulness of FTI and LSI and further study is warranted to verify the benefit of LSI.

We verified the effect of LSI on the immediate success and the elimination of DCs. Whether higher LSI actually leads to better clinical outcomes of AF ablation is not clear.

Conclusion

LSI was capable of predicting gaps or DCs in PVI procedure. LSI ≈ 5.2 may be a suitable target for safe and effective lesion formation. Lower LSI ≈ 4.0 may be acceptable in the posterior wall, particularly in areas adjacent to the esophagus.

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Tables

Table 1. Characteristics of the patients

	n = 34
Age, years	68.4 ± 6.8
Male, n (%)	26 (76.5)
BMI, kg/m^2	24.7 ± 3.1
Time from first diagnosis of AF, months	52.6 ± 73.7
Paroxysmal AF, n (%)	18 (52.9)
Persistent AF, n (%)	16 (47.1)
Left ventricular ejection fraction, %	62.4 ± 12.4

Left atrial diameter, mm	40.3 ± 5.7
Left common pulmonary vein, n (%)	2 (5.9)
BNP, pg/mL	99.2 ± 132.0
eGFR, ml/min	62.8 ± 12.2
Medical history, n (%)	
Hypertension	22 (64.7)
Diabetes	7 (20.6)
Coronary artery disease	2 (5.9)
Stroke or TIA	4 (11.8)
Heart failure	6 (17.6)

CHA_2DS_2-VASc score	2.6 ± 1.7
Antiarrhythmic agents, n (%)	
Beta-blocker	12 (35.3)
Sodium-channel blocker	4 (11.8)
Amiodarone	0 (0)
Bepridil	0 (0)

Data presented as mean ± standard deviation unless indicated otherwise. BMI, body mass index; TIA, transient ischemic attack.

Table 2. Contact force parameters with and without gaps or dormant conductions

	Without Gap/DC n = 3,054	**With Gap/DC n = 41**	**p-value**
Max contact force (CF), g	35.8 ± 16.8	29.8 ± 12.3	0.023
Average CF, g	16.1 ± 7.1	13.0 ± 4.9	0.005
RF duration, seconds	15.4 ± 9.2	11.3 ± 4.6	0.005
Force time integral (FTI), g・s	232.4 ± 121.4	140.5 ± 54.5	<0.0001
Lesion size index (LSI)	4.7 ± 0.9	4.0 ± 0.6	<0.0001
Average power, W	28.2 ± 3.0	28.1 ± 3.6	0.818

DC, dormant conduction; RF, radiofrequency; *n*, number of ablation points.

Figures

Figure 1. Procedure of pulmonary vein antrum ablation. We performed extensive encircling pulmonary vein isolation. A: The ablation points (red) were automatically tagged on 3D map by AutoMark system. B: The size of ablation tags was set as 4 mm in diameter. To achieve the targeted inter-lesion distance (ILD) of 3 to 4 mm, we intended to ablate so that a new tag overlapped with the former one.

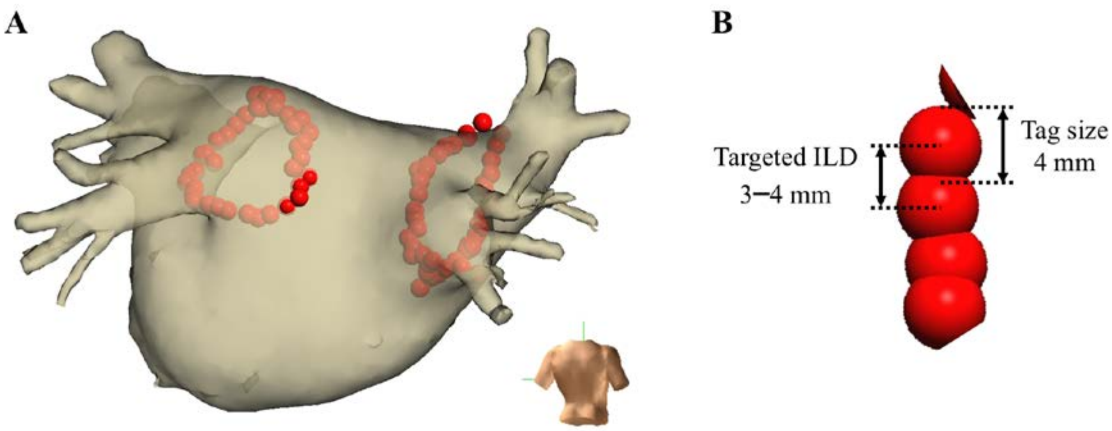

Figure 2. Myocardial wall thickness of the left atrium under the ablation line measured with multidetector CT. RS-RF, right superior roof; RA-S, right anterior superior; RA-C, right anterior carina; RA-I, right anterior inferior; RI-BT, right inferior bottom; RP-I, right posterior inferior; RP-M, right posterior middle; RP-S, right posterior superior; LS-RF, left superior roof; LP-S, left posterior superior; LP-M, left posterior middle; LP-I, left posterior inferior; LI-BT, left inferior bottom; LL-IR, left lateral inferior ridge; LL-CR, left lateral carina ridge; LL-SR, left lateral superior ridge.

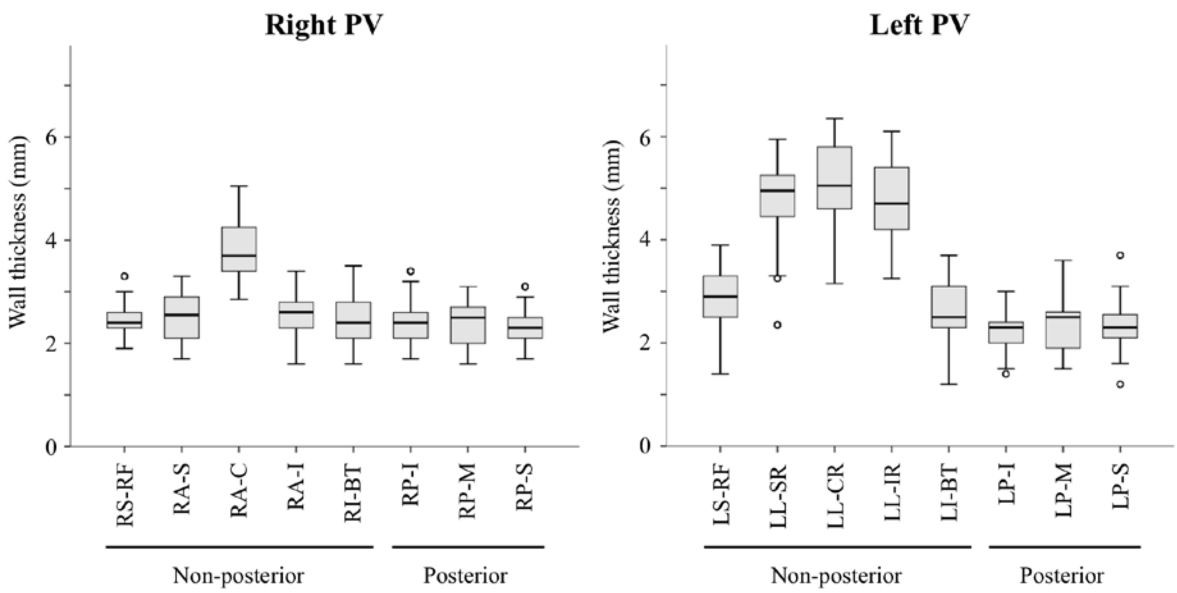

Figure 3. Distribution of FTI and LSI in each segment of the PVs. FTI, force–time integral; LSI, lesion size index.

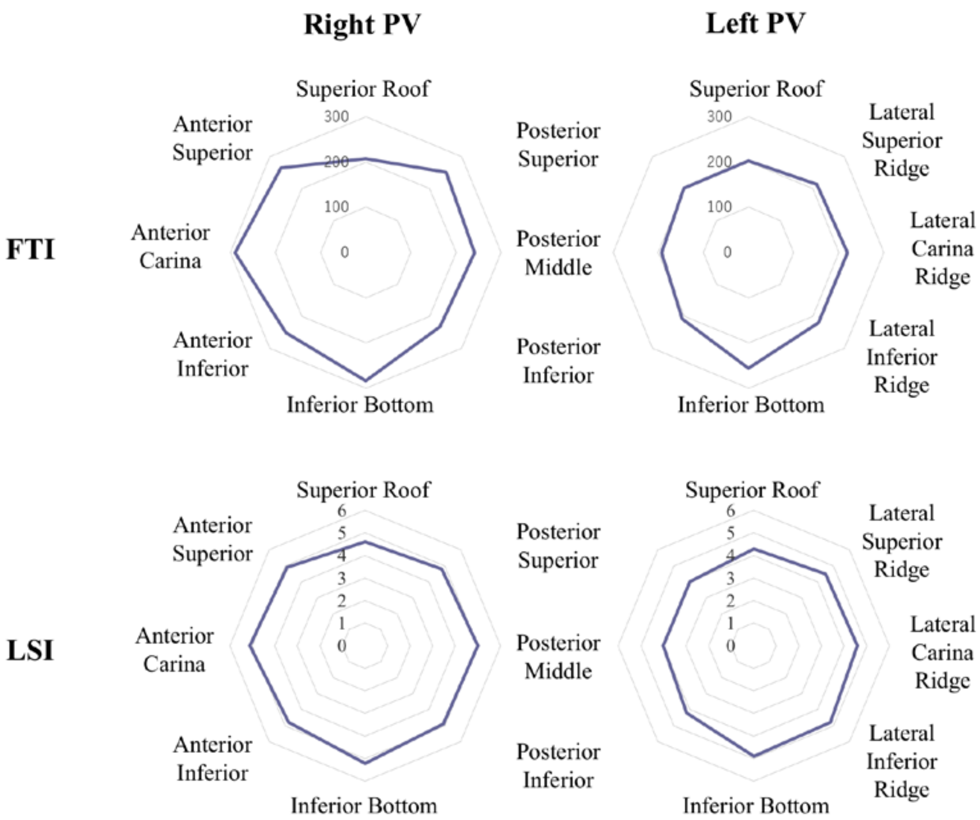

Figure 4 Correlation between LSI and FTI under different power settings. Power setting of 25 W (A), 30 W (B), 35 W (C). FTI, force–time integral; LSI, lesion size index.

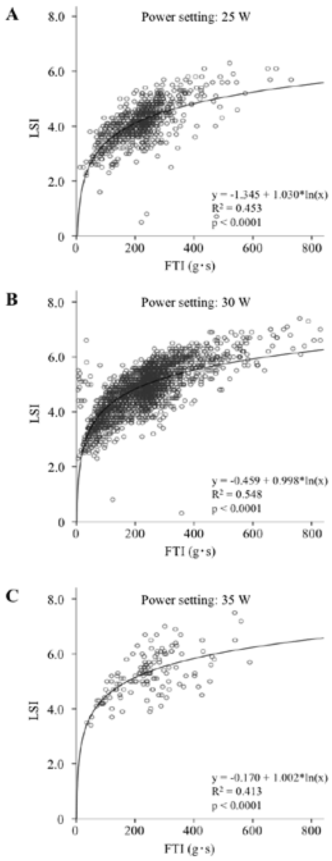

Figure 5. Distribution of LSI (Left) and FTI (Right) with and without gaps or dormant conduction. Gap indicates conduction gap after PV antrum ablation. FTI, force–time integral; LSI, lesion size index; DC, dormant conduction.

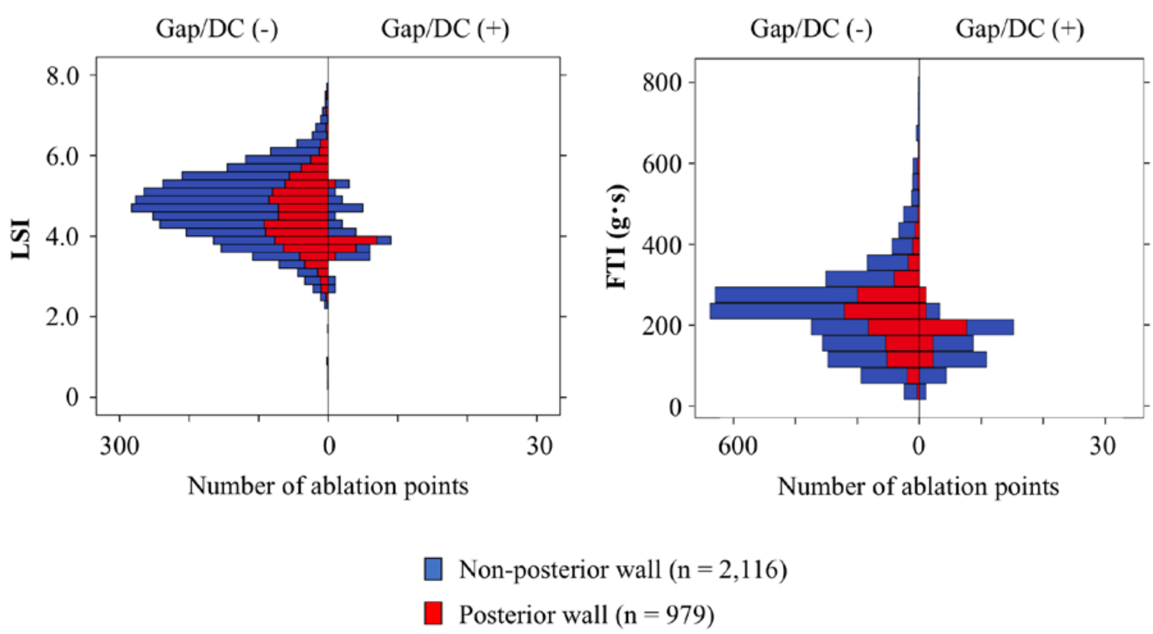

Figure 6. Receiver operating characteristic curve analysis to predict acute gap and dormant conduction (DC). The areas under the curve (AUC) for FTI and LSI for all points were not significantly different; 0.774 (95% CI 0.730–0.819) and 0.724 (95% CI 0.659–0.790), respectively (p = 0.21). Average CF, max CF, and duration of RF had AUCs of 0.641 (95% CI 0.562–0.721), 0.609 (95% CI 0.526–0.692), and 0.687(95% CI 0.606–0.768), respectively. The optimal LSI threshold to predict gaps or DCs was 4.05 (sensitivity 63.4%; specificity 76.3%). LSI threshold of <5.25 predicted gap or DC formation with a high sensitivity (sensitivity 97.6%; specificity 25.7%). FTI, force–time integral; LSI, lesion size index; CF, contact force; RF, radiofrequency.